

Lecture 8

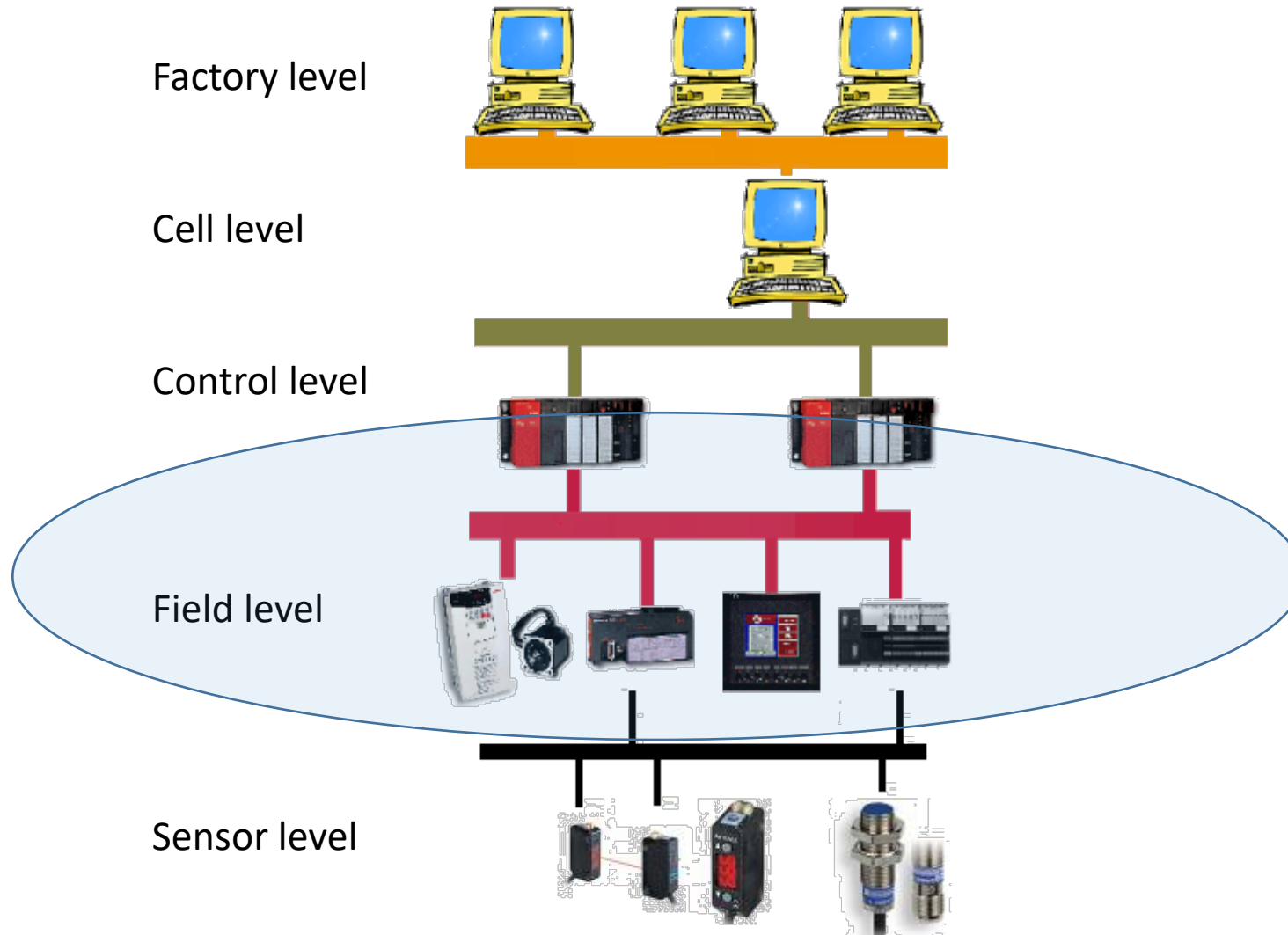
Fieldbus communication

- Overview of Field Bus standards on the market
- EMC issues, Potential equalization
- Trouble shouting

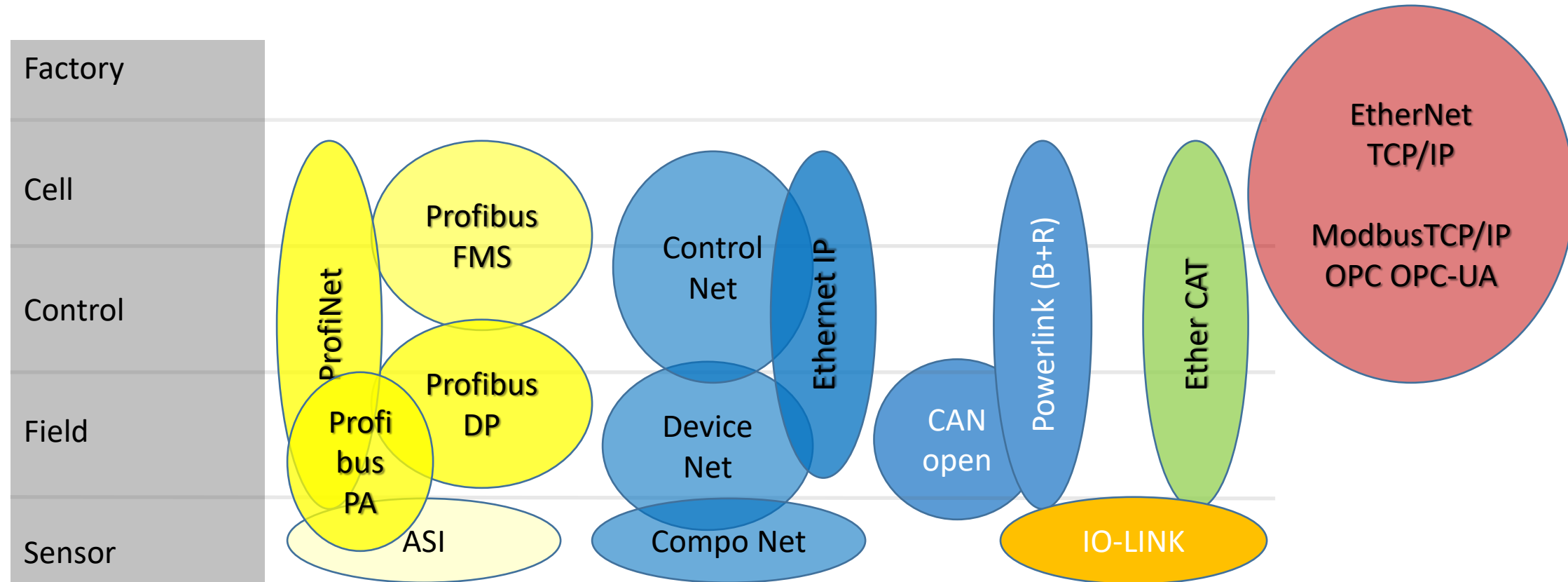
Field Bus Standards on the market

Operation methods
Comparations
strengths and weaknesses

Communication levels in a factory

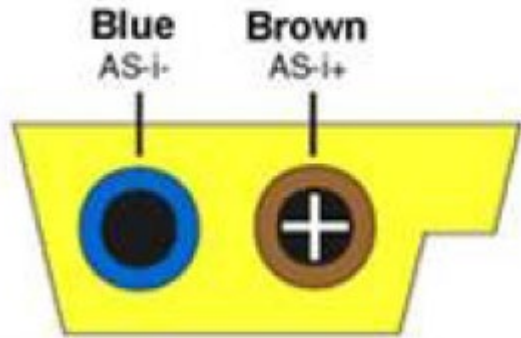


Communication standards divided in levels





Actuator Sensor Interface



- Developed by the sensor manufacturer (1991)
- Handled by AS-International
- Max 31 slaves max 4 input + 4 outputs
- Free topology due to slow communication rate, the cycle time of 5ms
- No terminal resistors are needed (Slow speed)
- Power supply and communication on the same wirings (Special Power supply)
- The address of each unit is programmed by a special device
- Automatic addressing on a replaced module



ASI safety

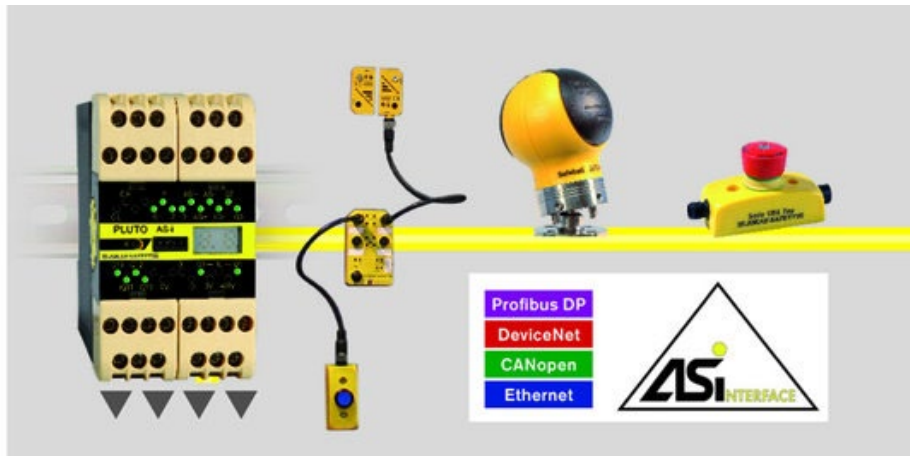


ABB Jokab Safety Pluto

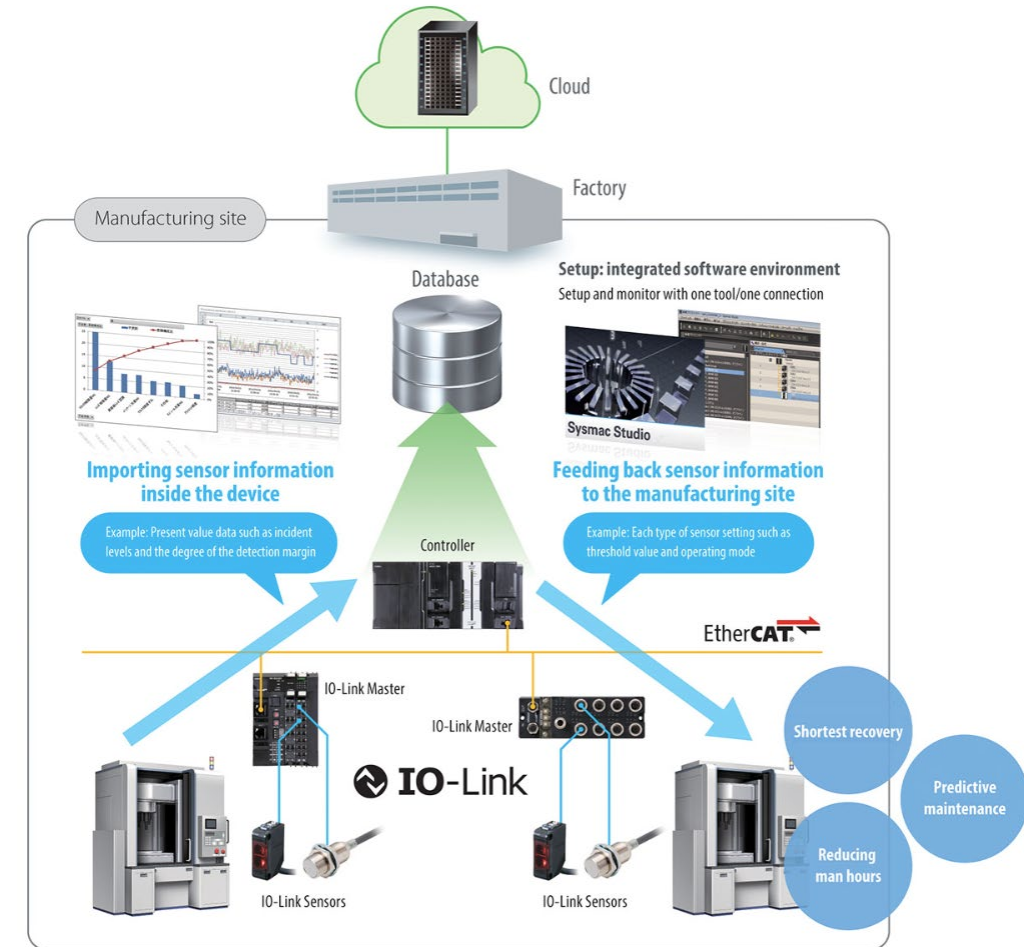


Telemecanique Harmony

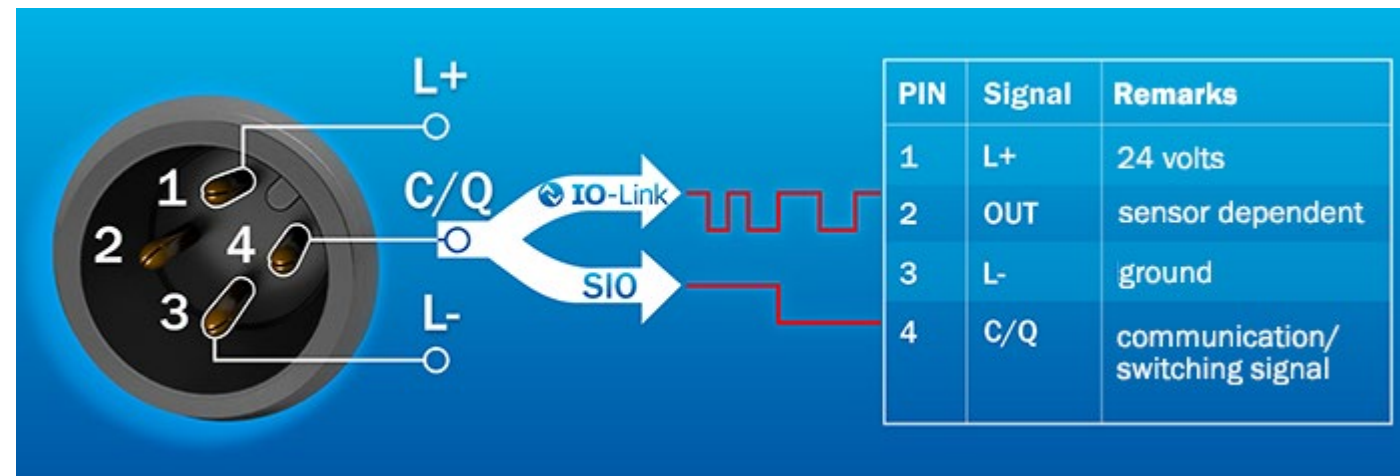
IO-Link

I/O-Link (IEC 61131-9)

- The first worldwide standardized, digital, point-to-point serial communication protocol (IEC 61131-9) for sensors and actuators.
- It acts as the "last meter" of communication, connecting smart field devices to the automation system.
- Not a Fieldbus: It is a point-to-point connection that works under existing fieldbus systems (PROFINET, Ethernet/IP, EtherCAT, etc.), not a replacement for them.



IO-Link



<https://www.sick.com/se/sv/io-link-allmaent-och-teknik/w/io-link-basics-and-technology/>

- Sensor and Actuator Point-to-Point Communication
- Cable length max 20 meters, Standard M12 connector
- Two modes
 - IO-Link: Analog and Status Information (Serial Communication)
 - SIO: Boolean True/False, Common 24V sensors and actuators



Controller Area Network

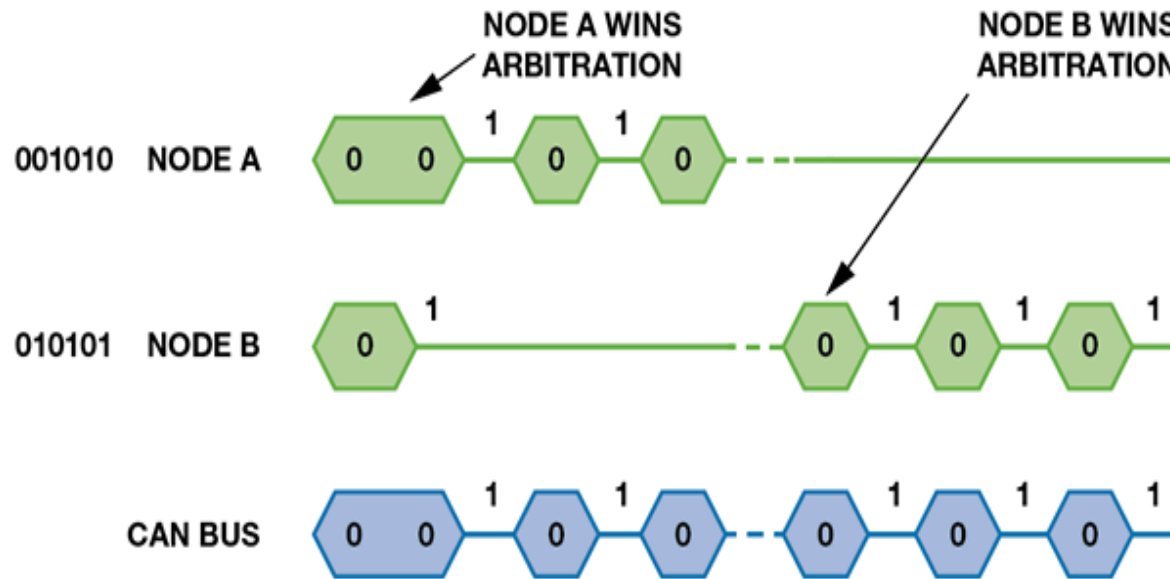
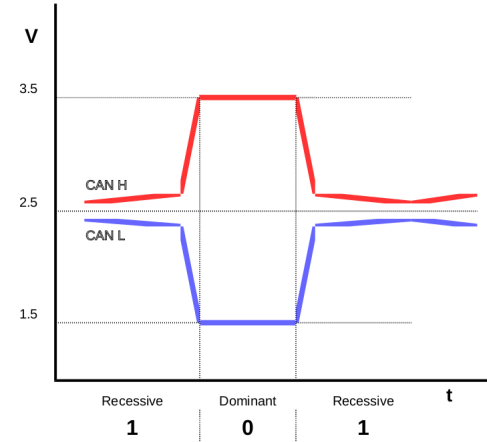
- CAN was developed in 1983 by Bosch for cars Available on the market in 1987
- Simple and cheap hardware (CAN is often integrated in small and cheap CPUs)
- High safety (used in antilock brake systems (ABS))
- Good on self-diagnostics
- Fast event-based multi-master system
- Max baud rate 1Mbit/s and max 8*8 bits (8-bytes) per telegram
- Bad at transferring big data
- CAN specify only the OSI layer 1 and 2
 - Layer 7 is needed as an add-on

Arbitration of Dominant bit (0)

The identifier determines the priority of the message

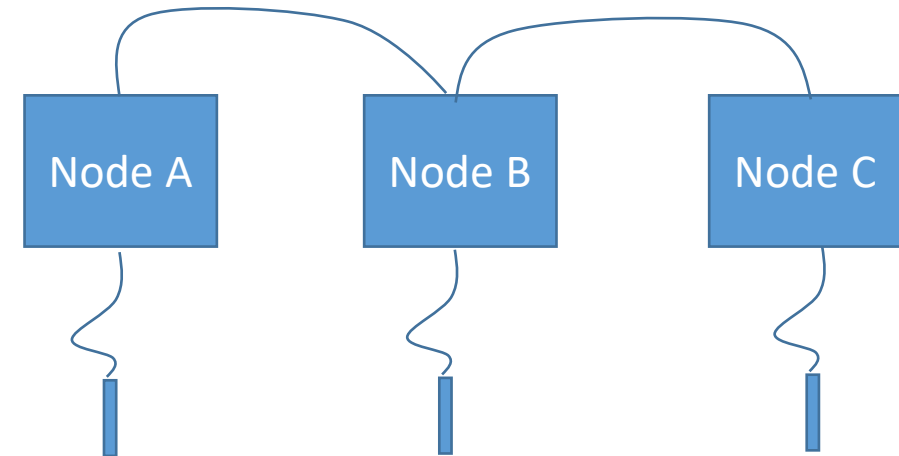
Lower identifier -> higher priority

High-priority messages will always arrive on time without delay due to Arbitration for a dominant bit



Arbitration! The dominant bit in the ID will win

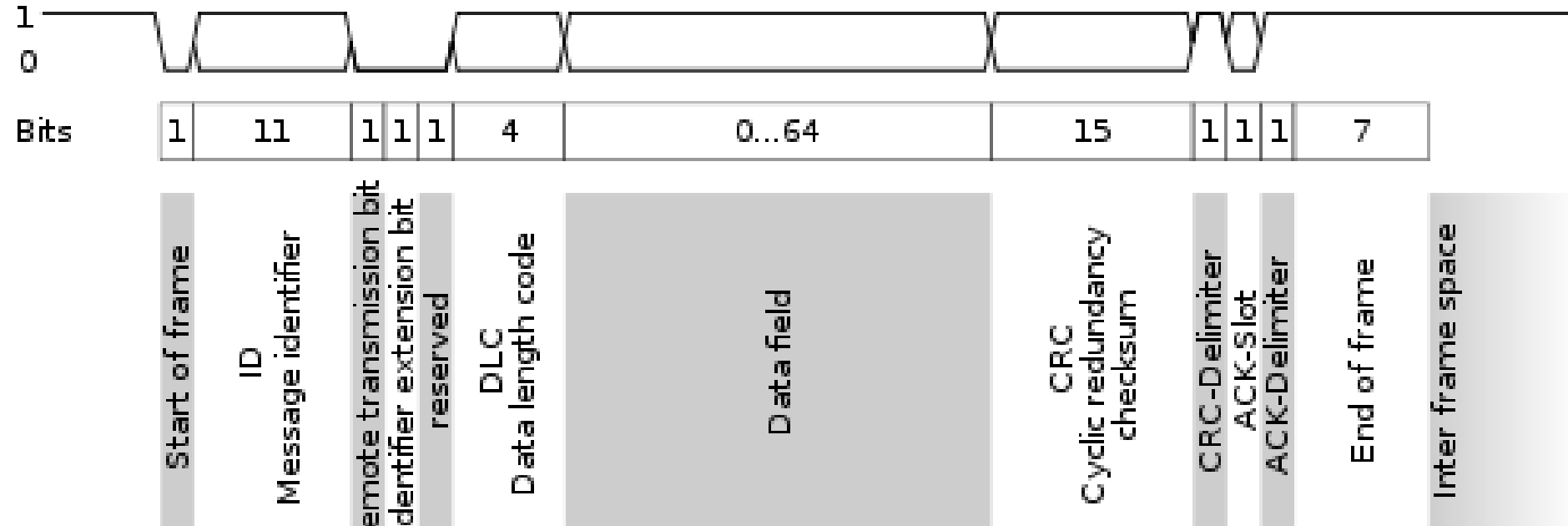
Multi-master system



Sensors connected to the nodes

http://www.analog.com/library/analogdialogue/archives/48-09/CAN_bit_timing.html

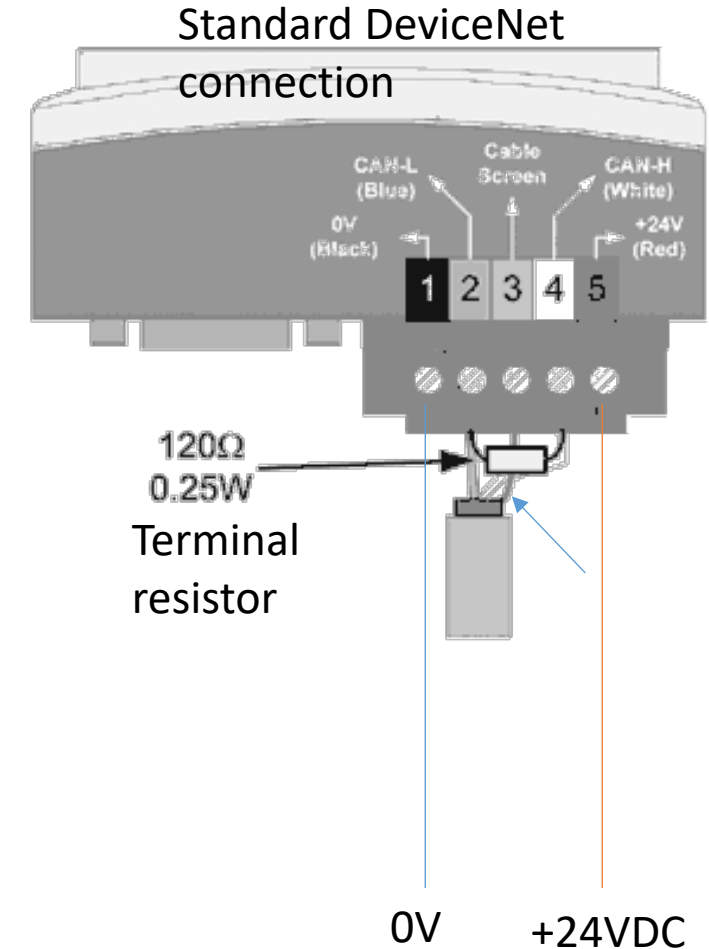
CAN Telegram structure





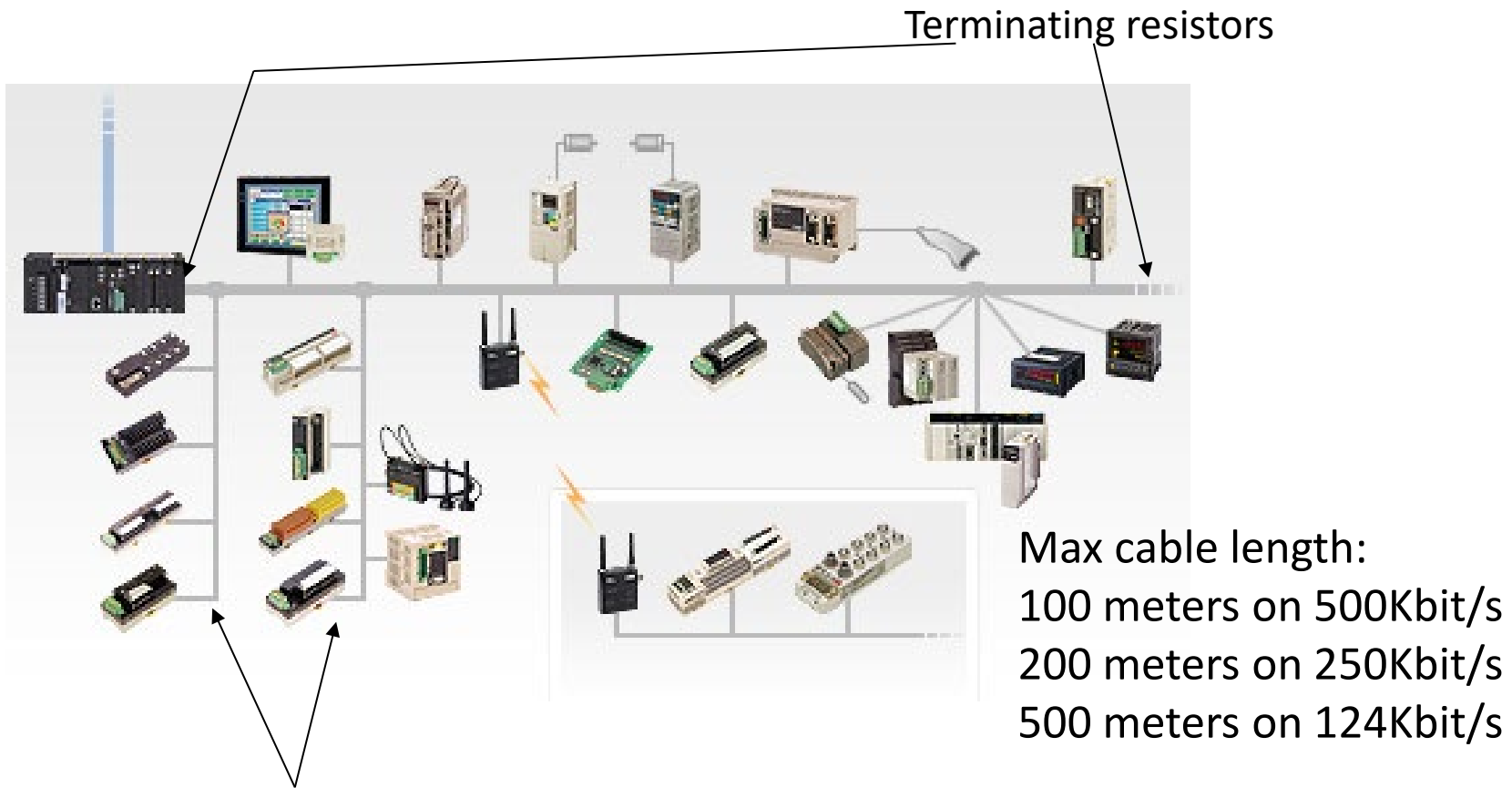
DeviceNet CAN for the industry

- DeviceNet (Layer 7) using CAN (Layers 1 and 2)
- Developed by Allen-Bradley (Rockwell Automation) for industry applications
- Managed by ODVA (Open DeviceNet Vendor Association)
- Open standard based on CIP (Common Industrial Protocol)
- Max 63 slave nodes and 1 master, all nodes have a unique address set by dip switches





DeviceNet/Can topology



CANopen

- Developed by “CAN in automation” (CiA)
- CAN version 2 max 127 nodes
- Time synchronized telegrams for coordinated servo axes
 - Often used for synchronized servo drives
 - CiA available in EtherCAT and PowerLink on Ethernet hardware
- Except for industrial applications, CANopen is often implemented in work vehicles and agricultural machinery such as cranes, excavators, harvesters, ...



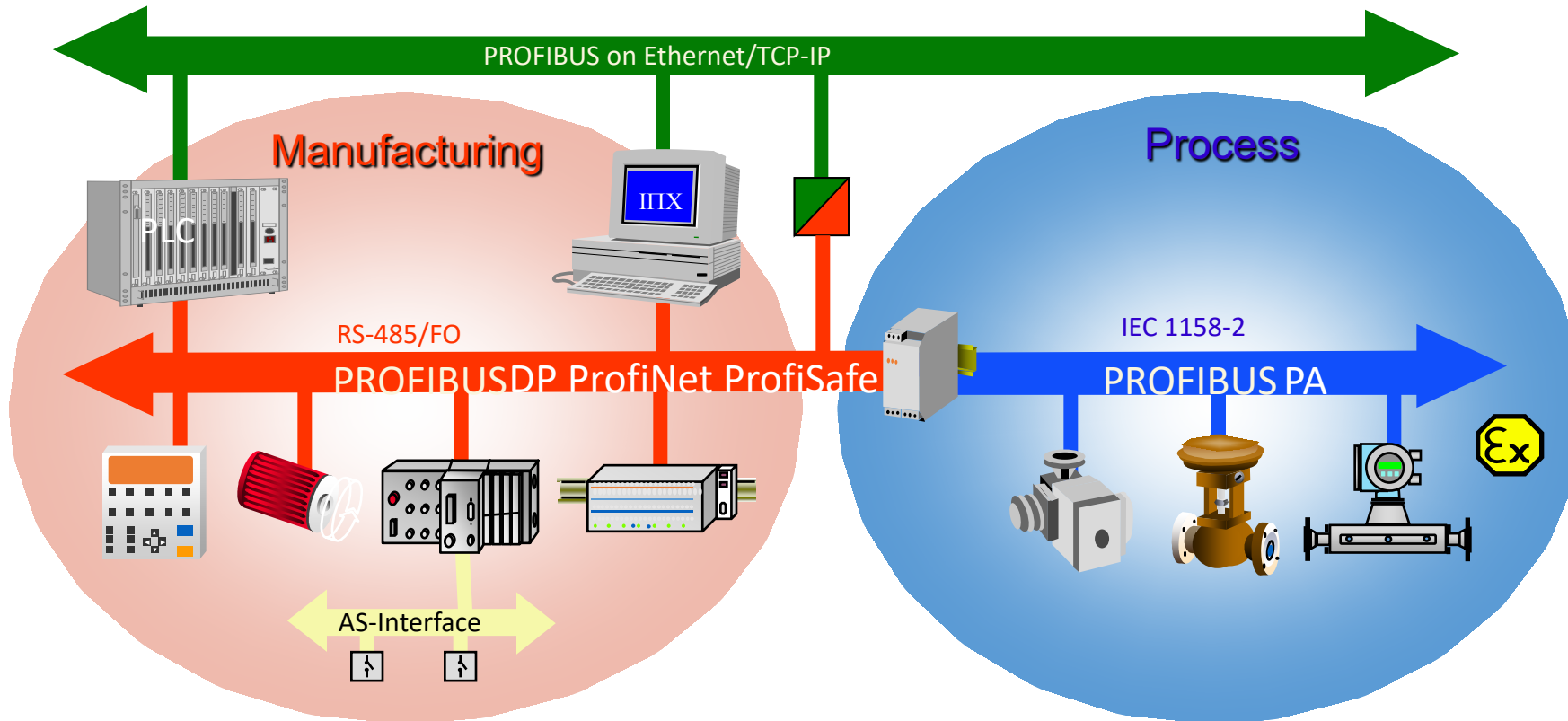
Profibus (Process Field BUS)

Process Field Bus developed by Siemens and managed of PROFIBUS INTERNATIONAL (PI)
Standard IEC 61158 resp. EN 50170
Introduced 1993

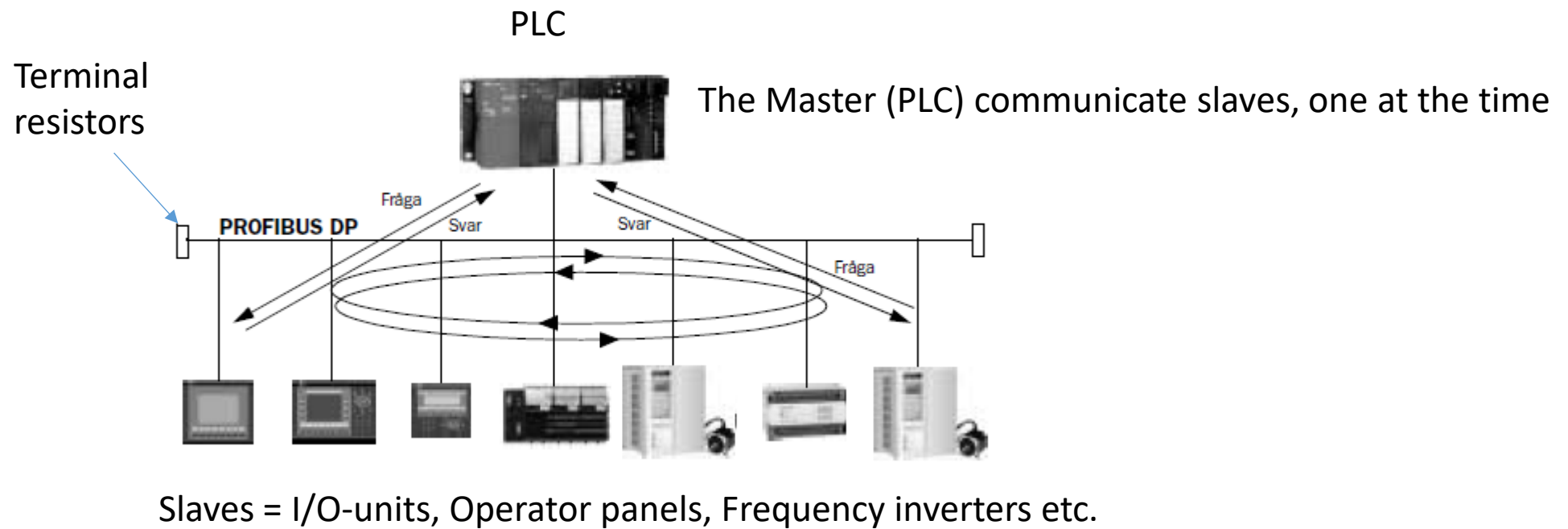
The family of Profibus:

- **Profibus FMS**. (Field Message Specification) intended for cell level
- **Profibus DP** (Decentral Perfori) intended for field level
- **Profibus PA** (Process Automation) Process industry applications in explosion-classified areas, communication and power supply in same pair of wires
- **ProfiNET** Profibus on Ethernet
- **ProfiSafe** ProfiNET with a safe profile

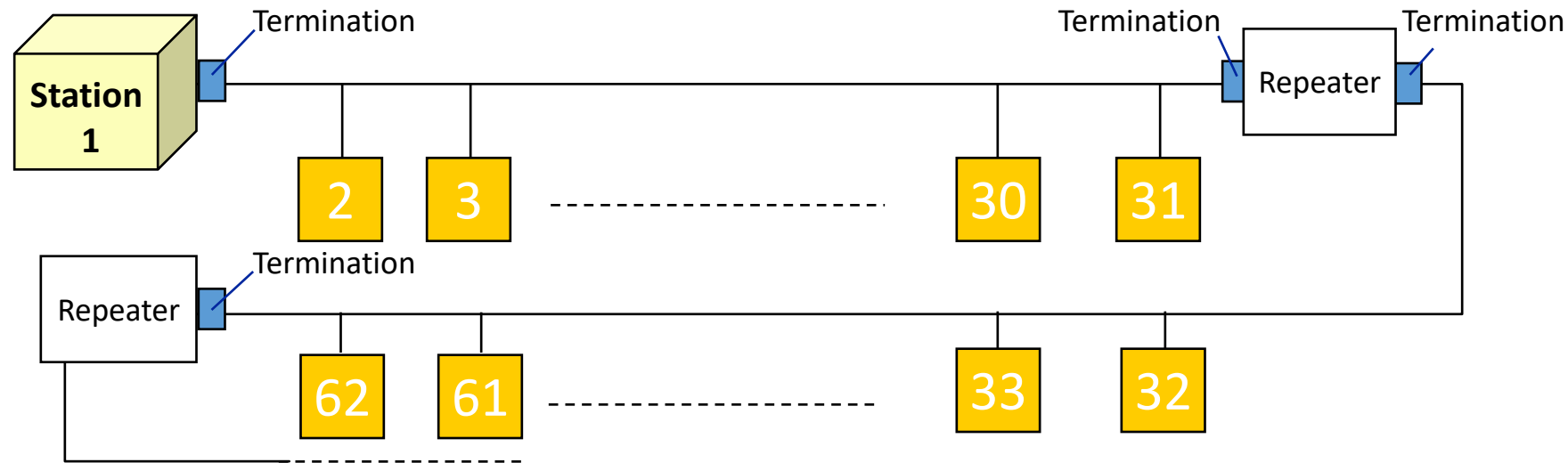
The Profibus family



Profibus DP V0 polling operating method



Net work topology



- 32 nodes in each segment without a repeater.
Max 126 nodes with 9 repeaters
- A repeater is often used for changeable robot tools when an I/O module is located on the robot tool

One GSD File to each slave node

- Each connected slave must be configured in the master by a configuration tool.
- The parameters/options are described in the GSD files.
- The communication system (nodes, connections, and addresses) is described in the configuration tool

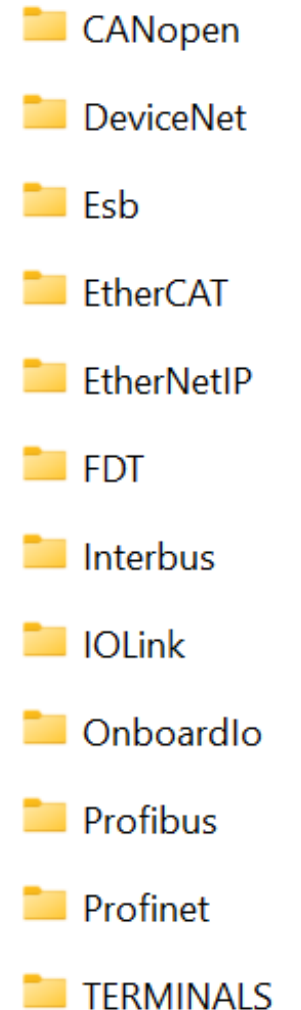


Configuration tool
Built-in TwinCAT



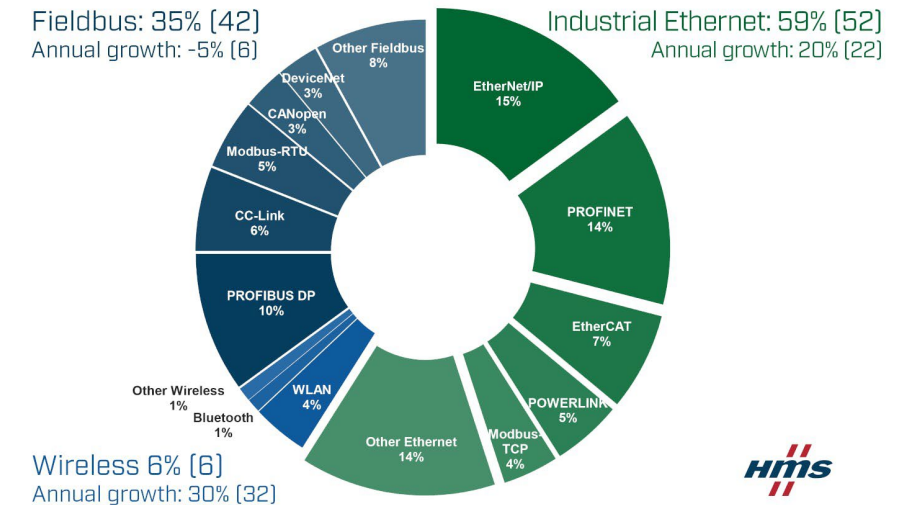
Config files in TwinCAT

- C:\Program Files (x86)\Beckhoff\TwinCAT\3.1\Config\Io\
- EtherCAT config files are in the .xml format

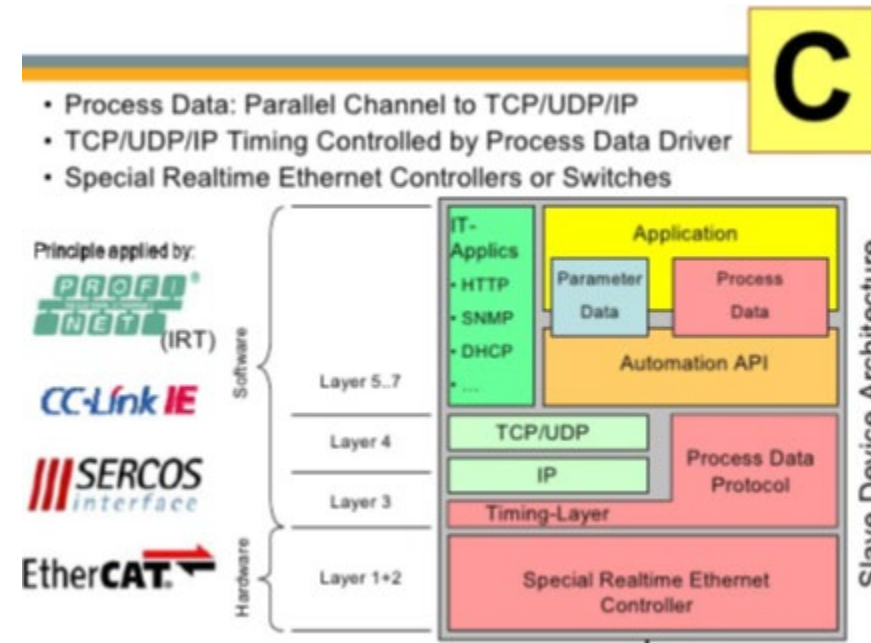
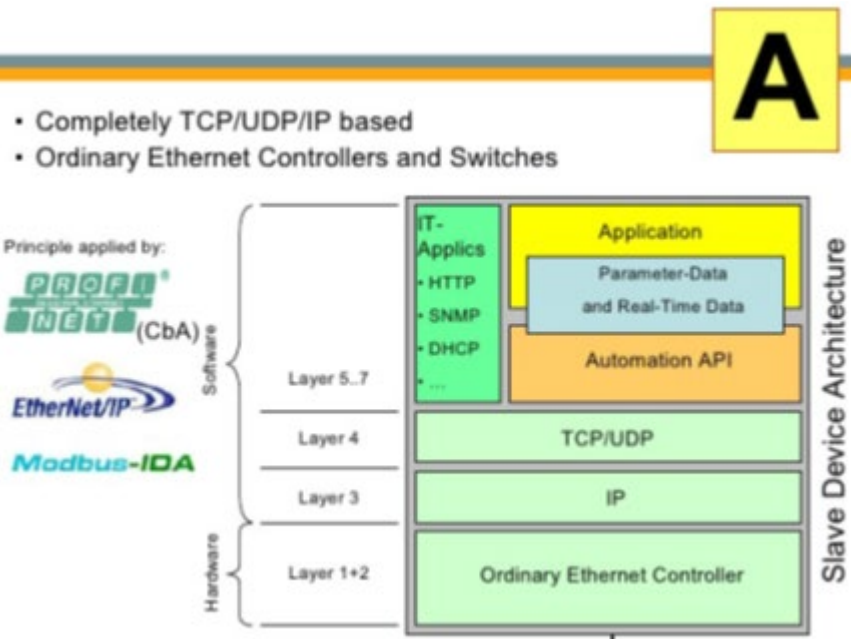


Ethernet-based fieldbuses

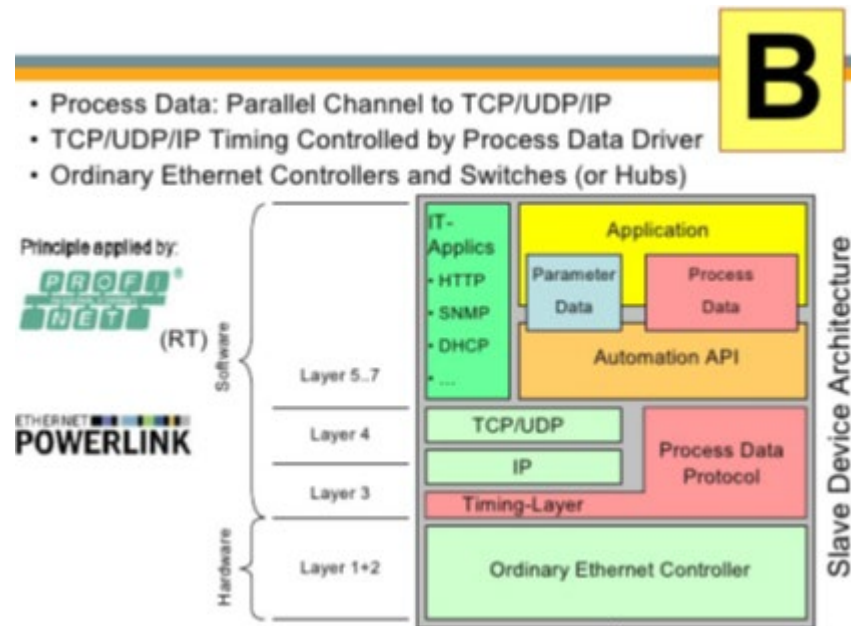
<https://www.novantaims.com/technology-blog/the-case-for-industrial-ethernet-ii-protocols/>



- **Ethernet IP** Devicenet (CIP) over Ethernet, Rockwell Automation
- **ProfiNet** three versions from non-real-time **CbA**, to **RT** and **IRT**, Siemens
- **EtherCAT** message frame passes through the nodes, Beckhoff
- **Powerlink** CANopen (CiA) over Ethernet, B&R Automation
- **ModBusTCP/IP** non-real-time RS232/485 on ethernet, Modicon



http://www.ethercat.org/2011/italy/download/04_Industrial_Ethernet_Technologies_IT_1102.pdf



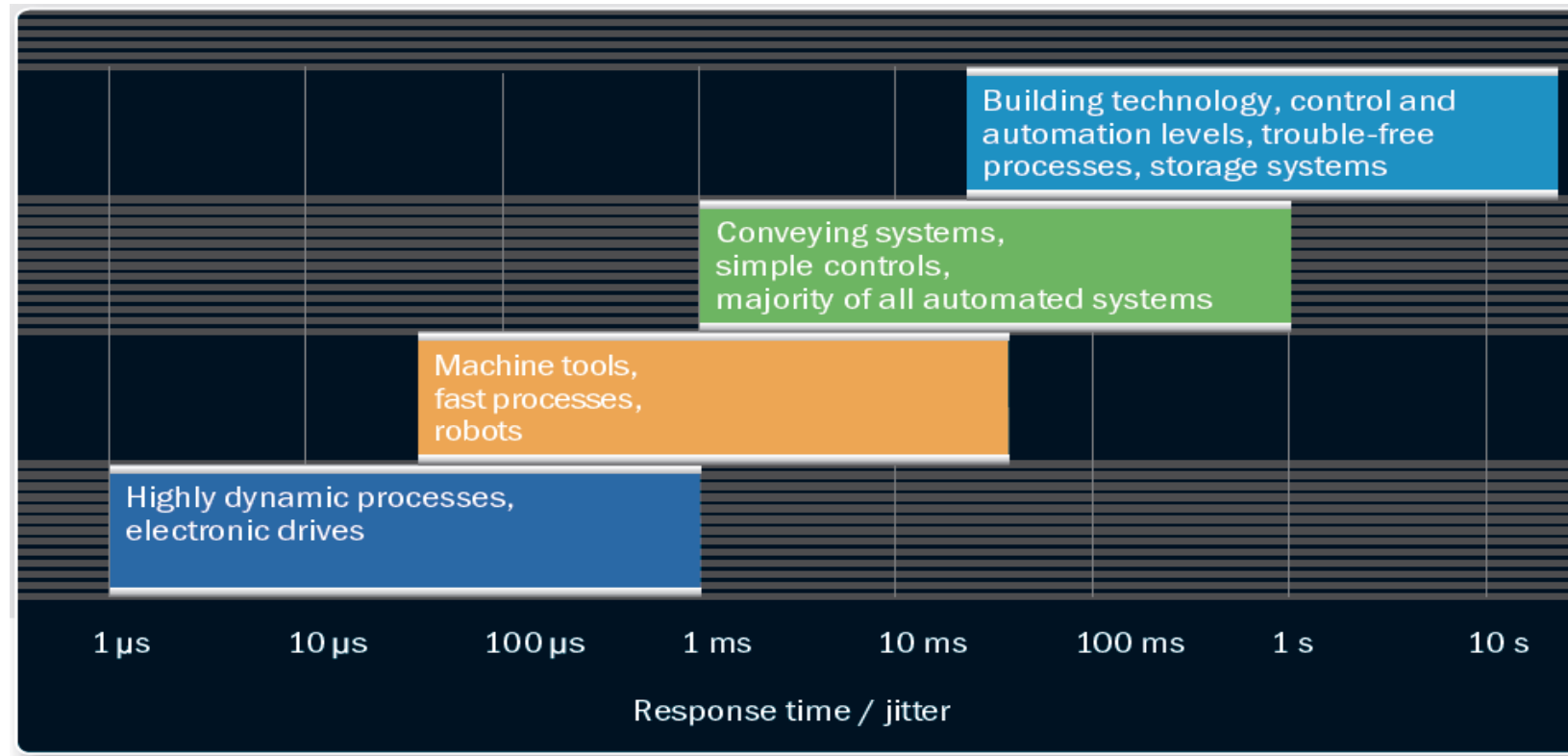
A: No real-time, unpredictable delays

B: Soft real-time; software solutions

C: Hard real-time; special Ethernet controller

All these Ethernet-based field buses can be combined with ordinary TCP-IP traffic

What is “real-time”? Defined response time?



Industrial Ethernet

Traditional fieldbuses

TCP-IP

Hard real time

Soft real time

EtherCAT[®] Overview

Ethernet for **C**ontrol **A**utomation **T**echnology (EtherCAT)

- Developed by Beckhoff Automation
- Today, an open standard, handled by **E**therCAT **T**echnology **G**roup (ETG)

“Hard” real-time performance:

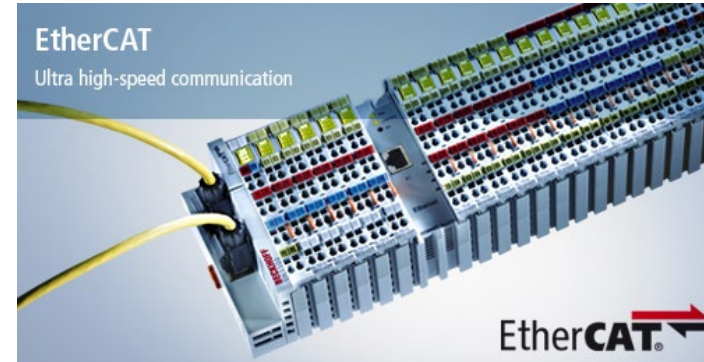
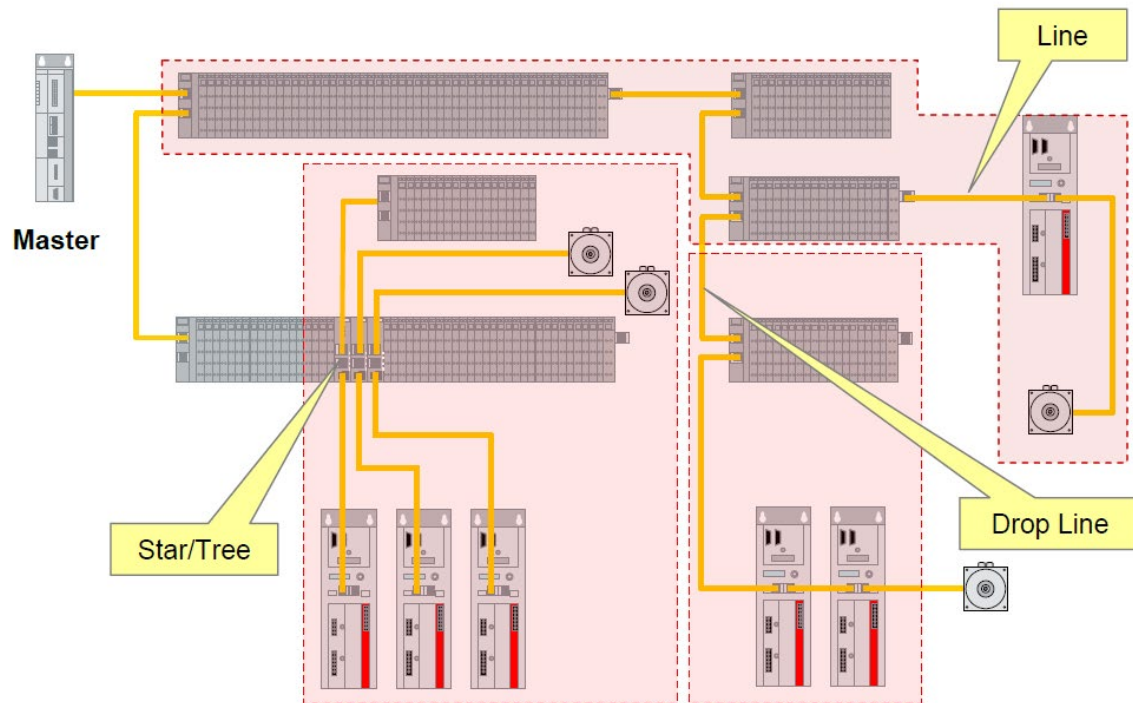
256 digital I/O in 11 μ s

1000 digital I/O distributed to 100 nodes in **30 μ s** = 0.03 ms

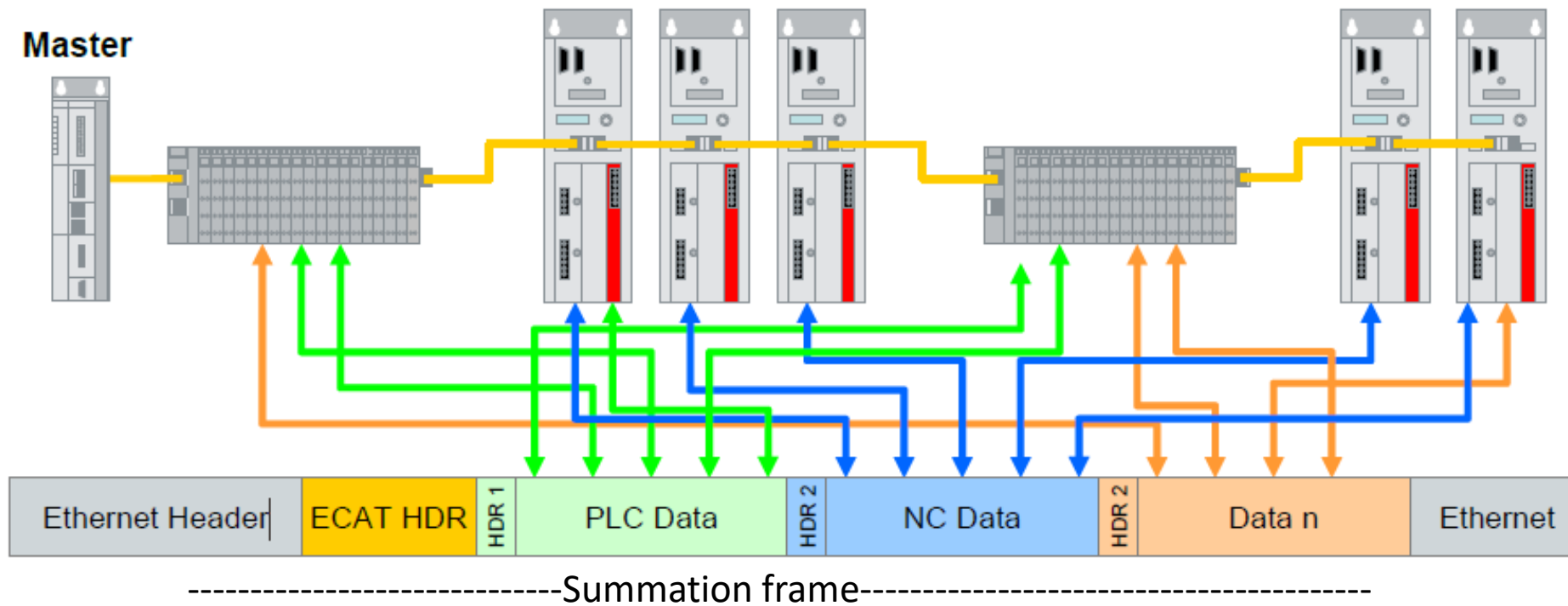
source **E**therCAT **T**echnology **G**roup ETG

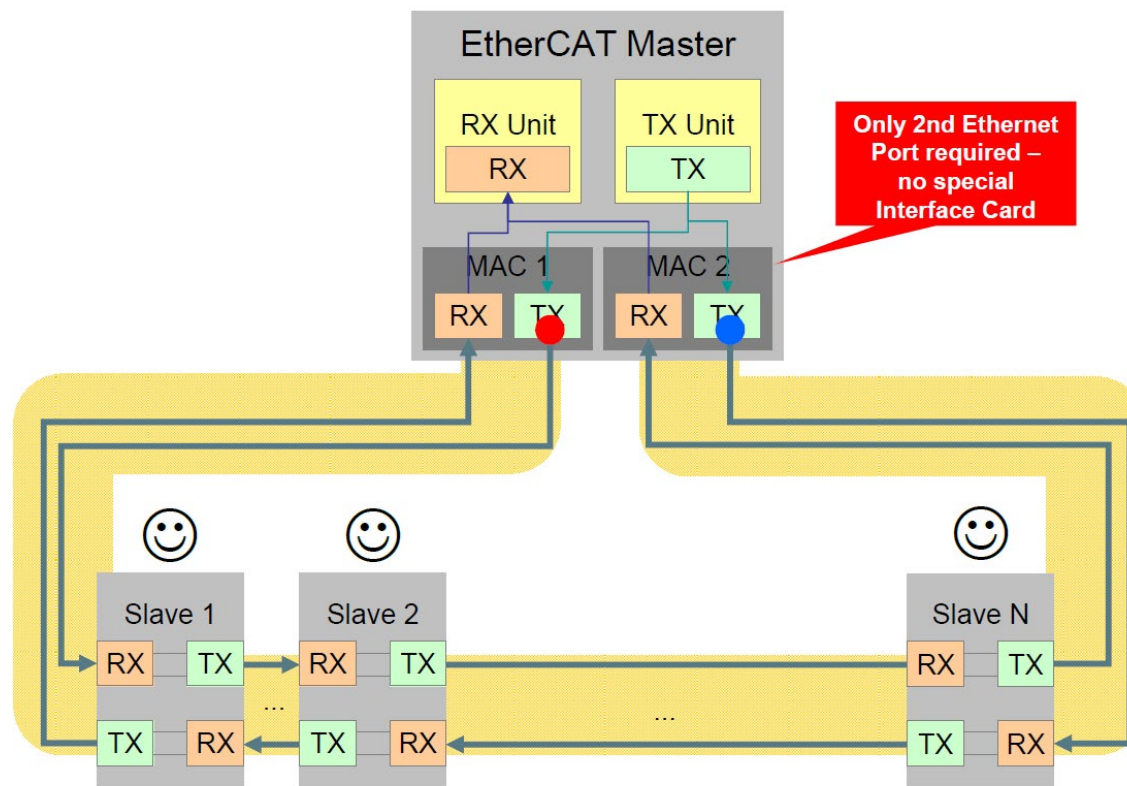
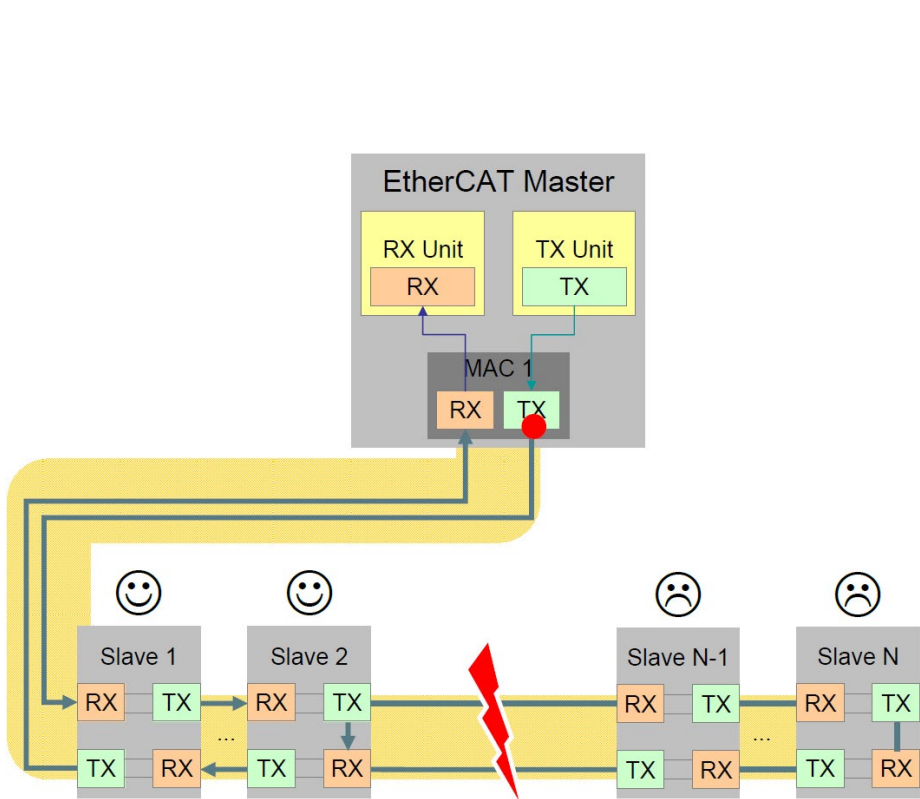
Physic layer and topology

Ethernet standard hardware connections and cables

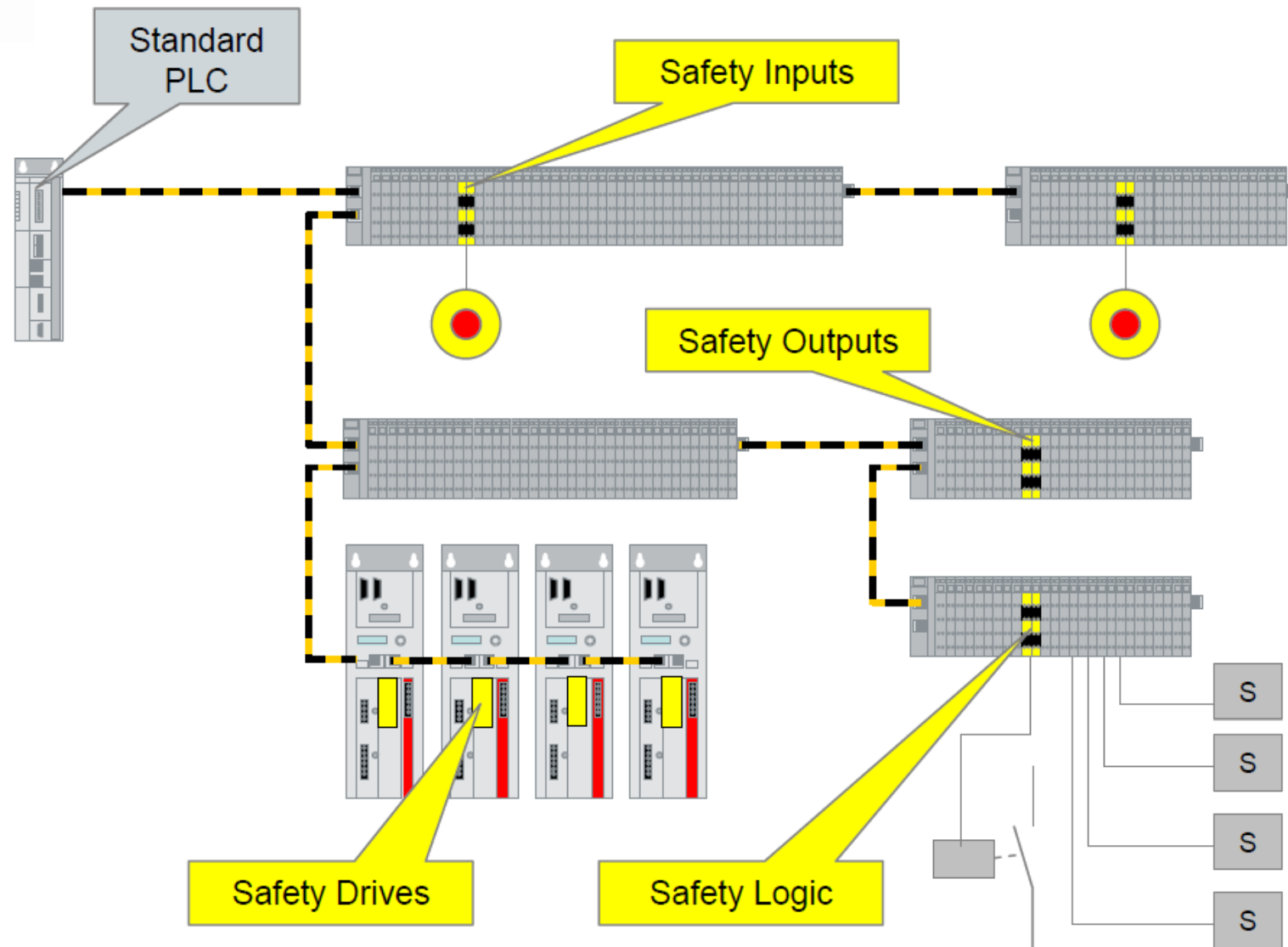


- Minimal protocol overhead via implicit addressing
- Optimized telegram structure for decentralized I/O
- Master clock for increased performance





EtherCat with safety profile



Anybus

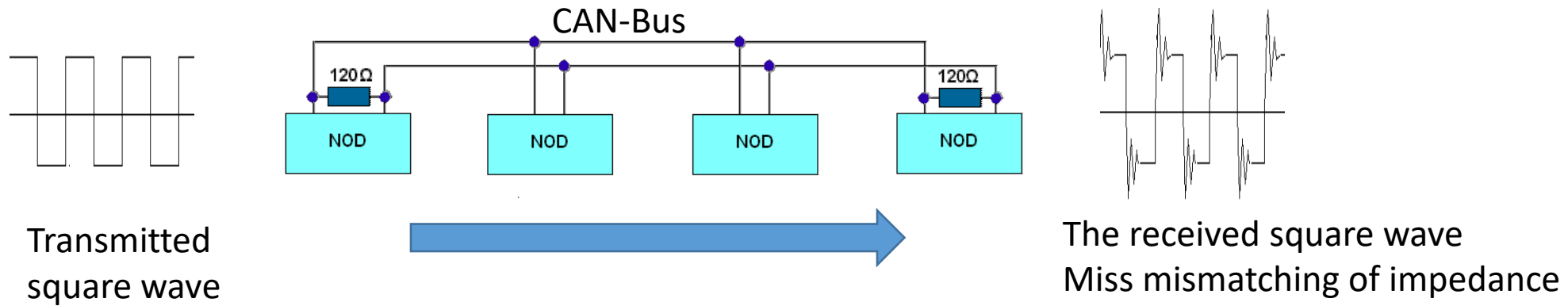
- Solves interconnection trouble
- Developed by HMS (www.hms.se)
- No one for safe technologies!



Connectivity solutions for



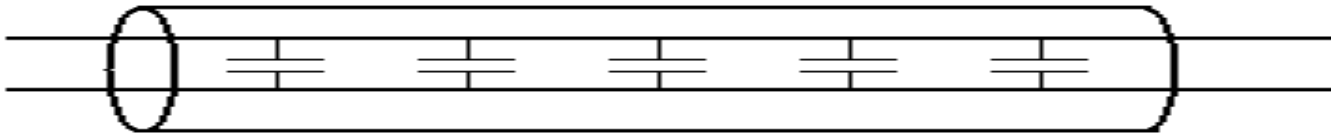
Why have termination resistors?



- Square pulses consists of a mix of high-frequency signals (Close to radio waves)
- Terminal resistances avoid the high-frequency signals from bouncing back again when the wire runs out
- A resistance of the same size as the cable's characteristic impedance acts as an extension of the cable
- Wrong resistance appears as "ringing waves"
The same thing happens in a connection of two different types of cable, squeezed cable, or in a bad connector

The length of the cable

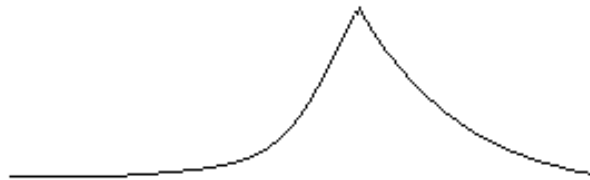
- Between the wires, there is a capacitance that increases for each meter
- Each conductor also has a resistance, which forms an LP-filter
- High communication speed (short pulses) implies shorter cable
- A good bus cable has a capacitance of 20pF/meter
- Each node adds capacitance, which means that nodes reduce the possible length



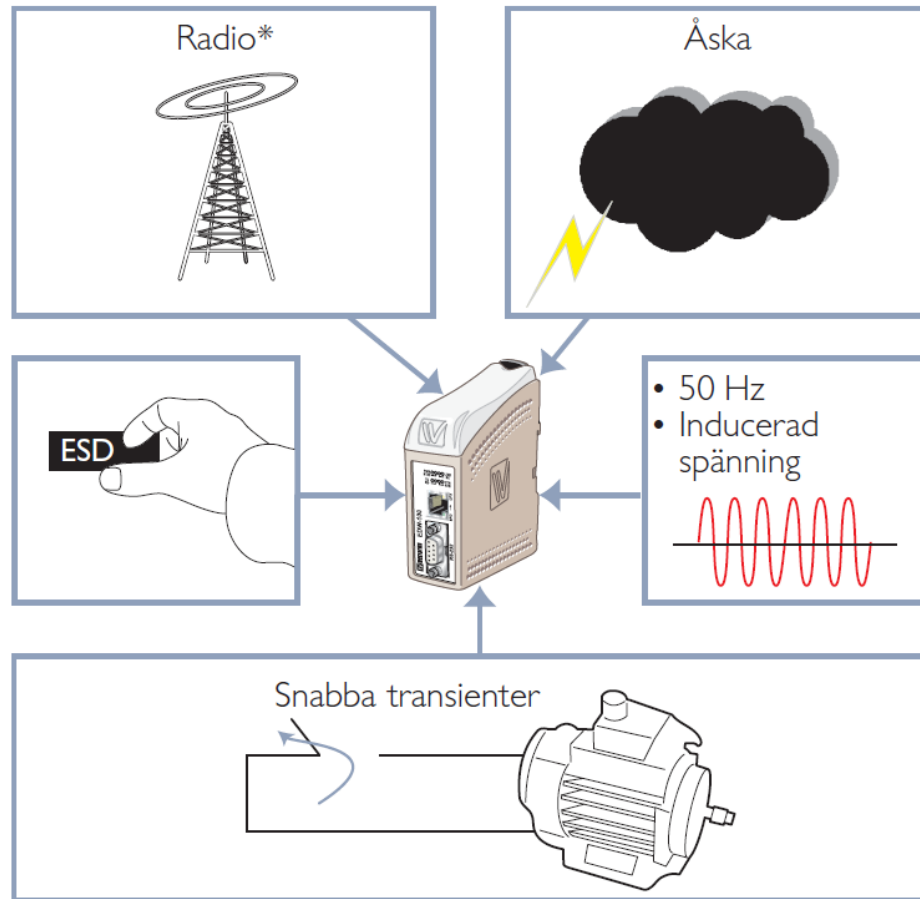
Ideal signal pulse



Signal that passed through a long cable



How to protect against interference?

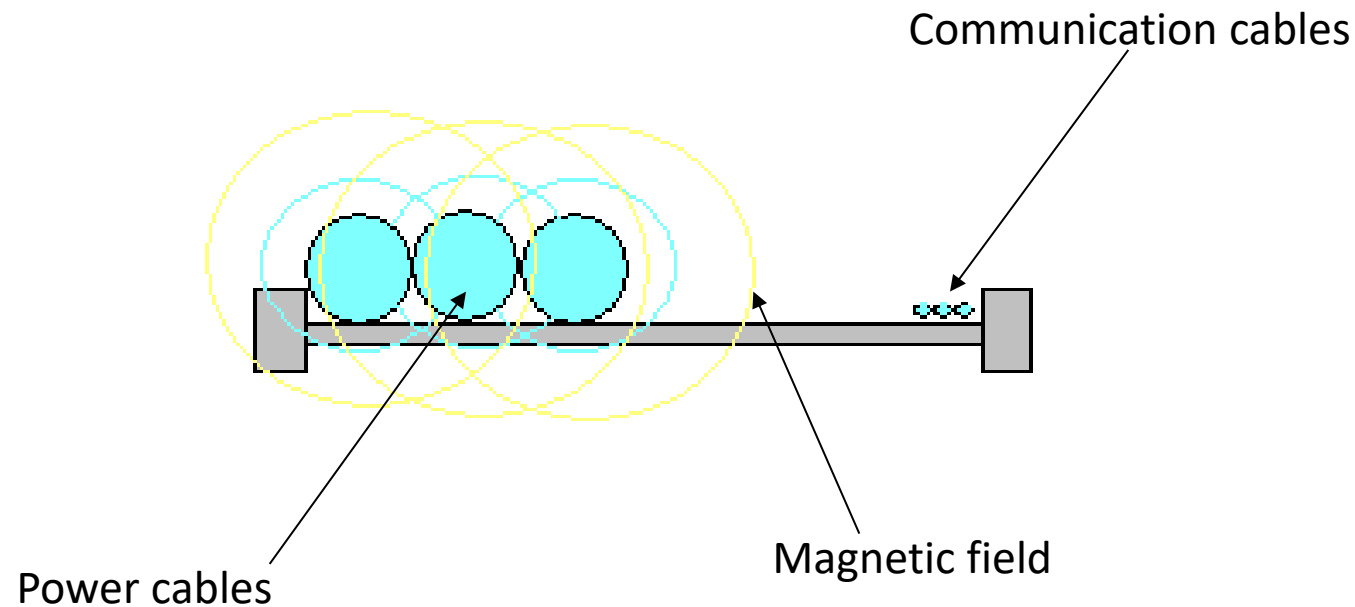


Interference from:

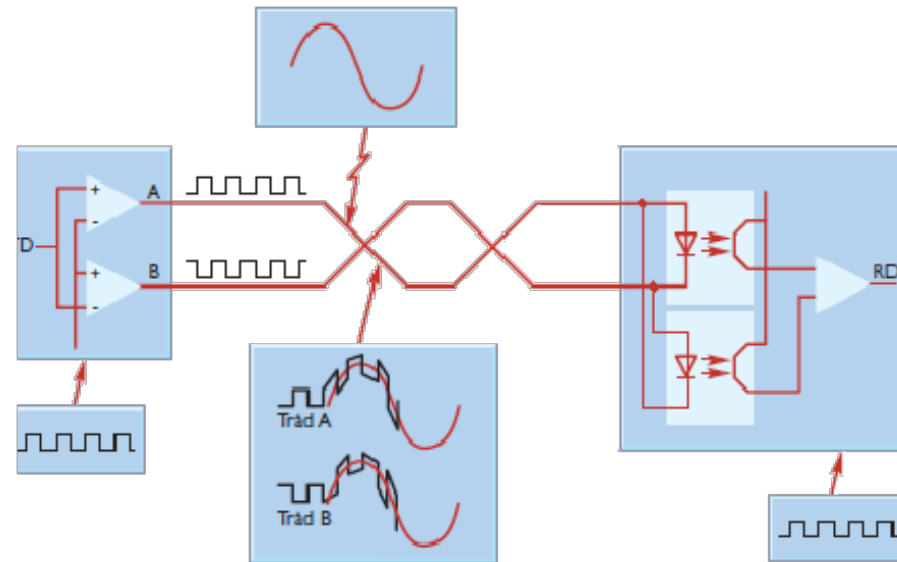
- Radio
- Thunder
- Induced voltage
- Transients
- Electro Static Discharge

Separate the communication cables and power cables!

- The protection of the screen becomes weaker and weaker on longer distances



Twisted pair wire and optic coupler



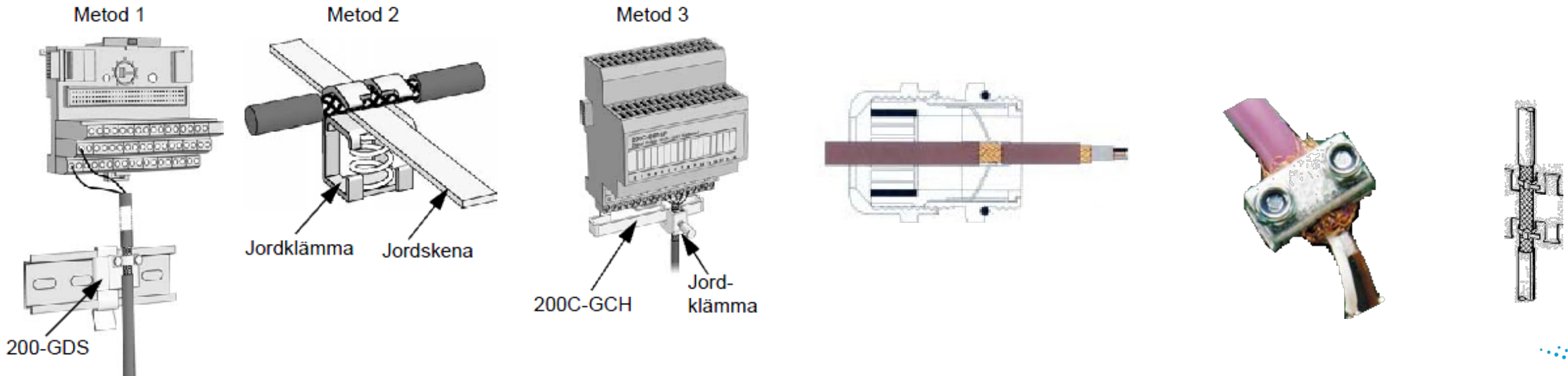
1. Data to the transmitter
2. B inverse to A
3. Incoming Interference (Induced voltage)
4. Interference overlaid with data
5. Fresh data to the receiver

Do not make antennas of the cable screen!

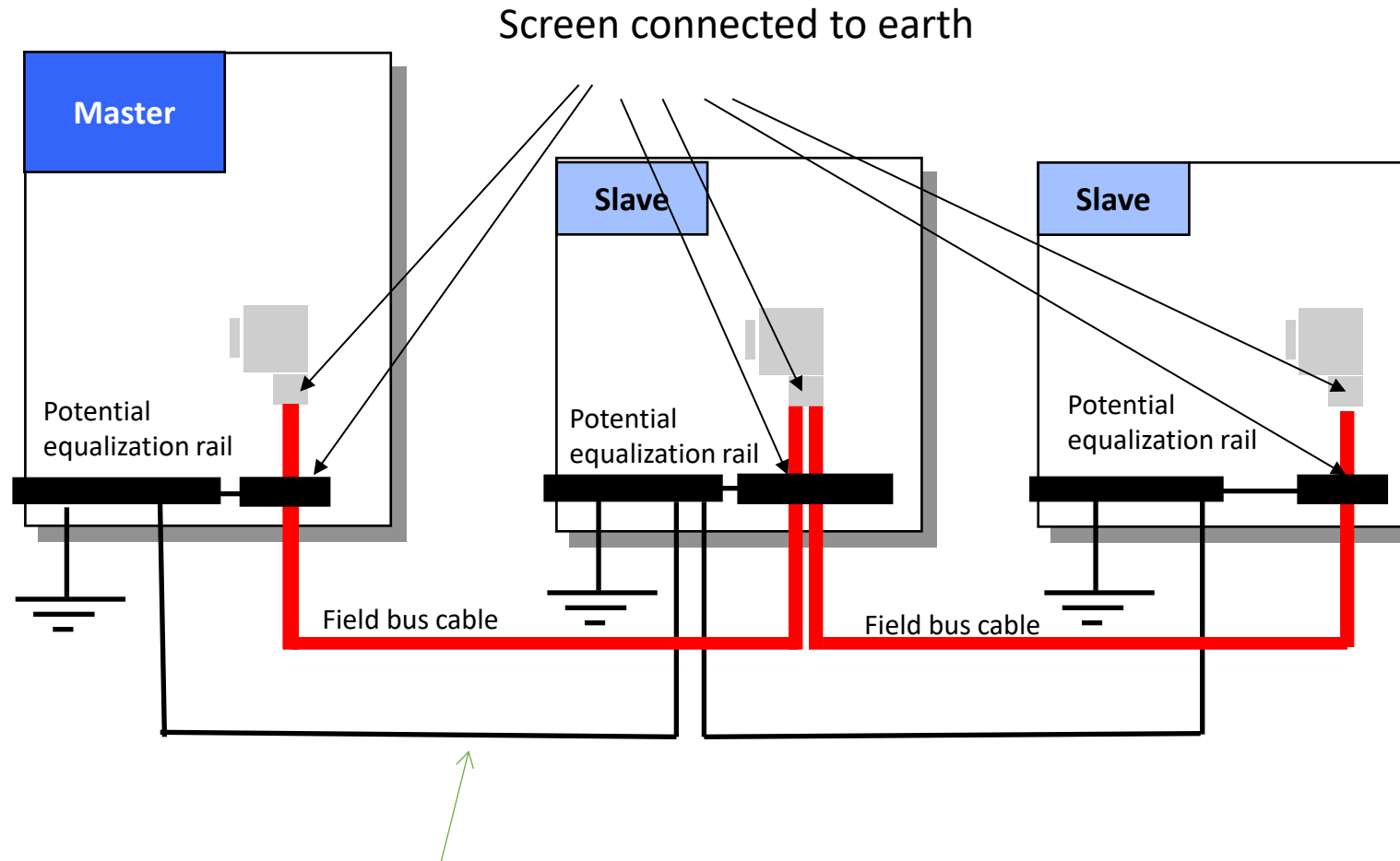


Properly grounded screen is important

- Combination of foil shield and braided screen protects against both high and low-frequency disturbances
- The screen must be connected to the contact's metal case, EMC cable screw, or earth clamp immediately when the cable goes into the cabinet
- If the cable goes into a galvanic separated zone, such ex-classified area.
 - Ground the screen via capacitor $\leq 10\text{nF}$ 1500V



Potential balancing and grounded cable screens



Potential equalization cable between all machine parts where there are nodes

Troubleshooting by dividing and survive

Switch to termination in the middle of the bus and see if the first half will start to locate the fault

